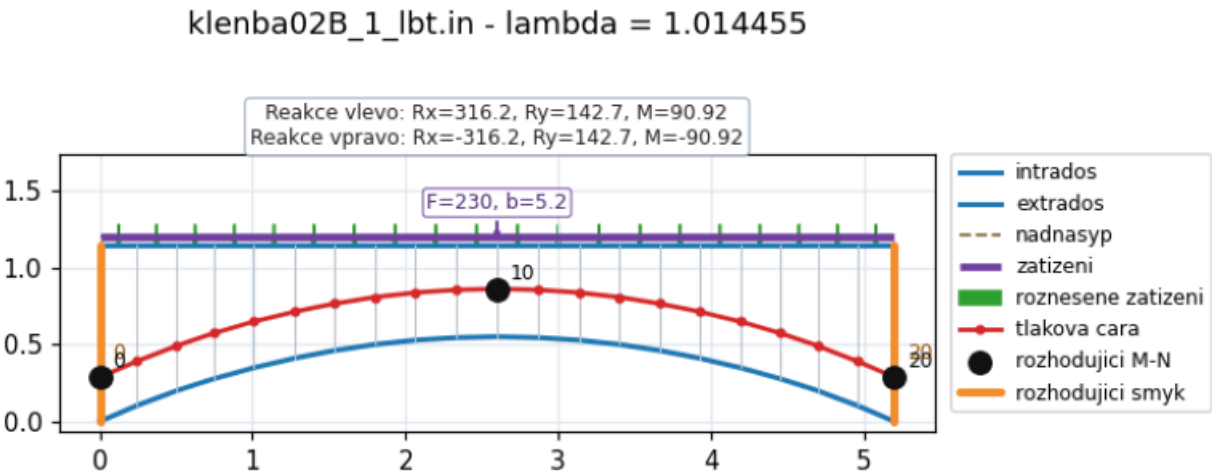


ArchLBT vypočetní protokol

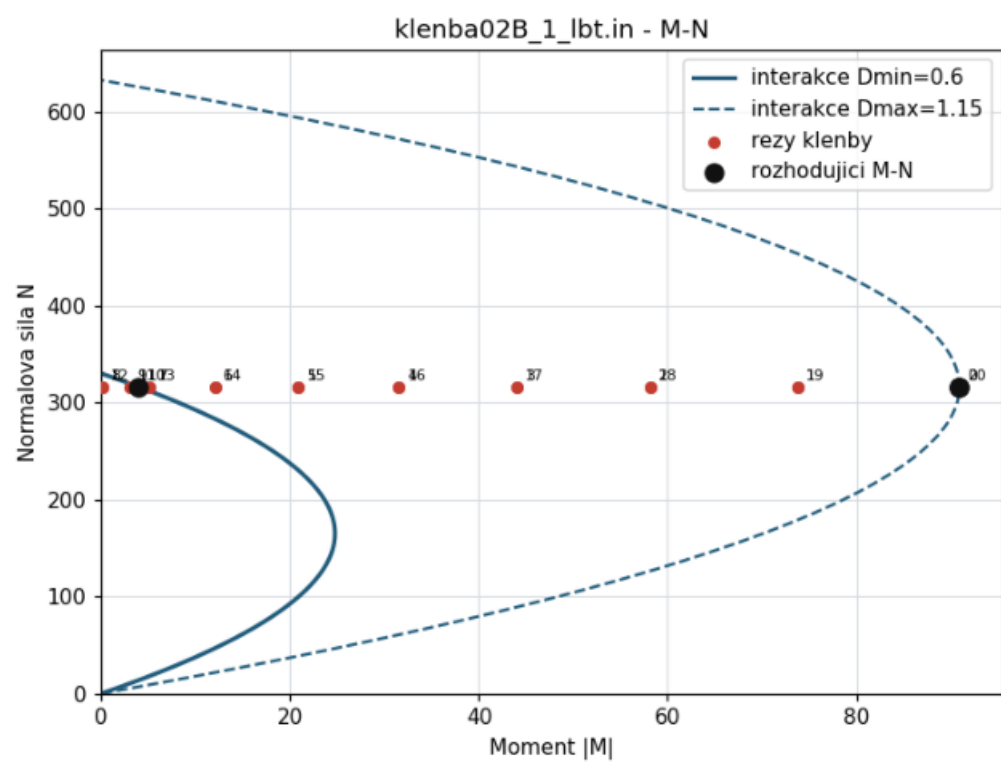
1) Zadání úlohy

```
# ArchLBT input
# Unit convention: use one consistent force-length system.
# Geometry is in length units, typical m. Unit weights are force/volume, typical kN/m3.
# Surface point_load force F is the total force on the loaded length, typical kN.
# Self-weight and fill loads are multiplied internally by arch_width; point_load F is not.
span = 5.2
rise = 0.55
geom_code = 1
# extrados_mode: normal = constant normal thickness, horizontal = flat level extrados.
extrados_mode = horizontal
thickness = 0.6
# arch_width is the out-of-plane width b of the analysed arch strip.
arch_width = 0.5
masonry_unit_weight = 24.3
block_count = 20
point_load = 2.6 230 5.2
fill_height = 0.05
fill_unit_weight = 21.76
q_code = 1
fill_spread_angle_deg = 0
friction_coefficient = 0.6
compression_strength = 1100
```

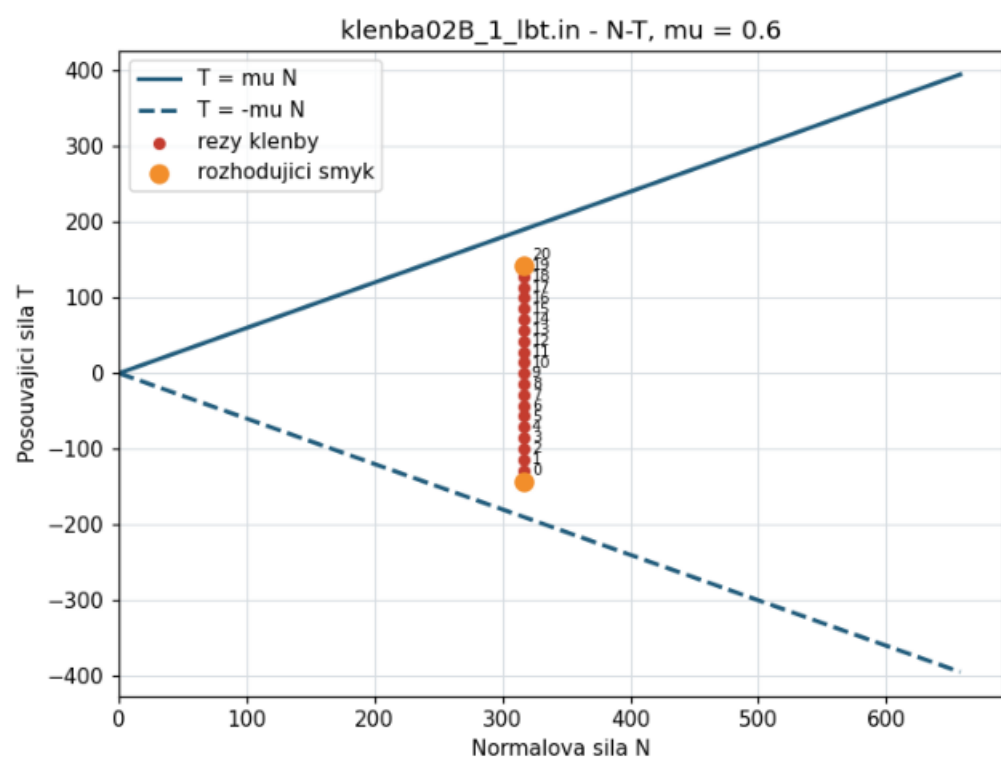
2) Obrazek Klenba



3) Graf M-N



4) Graf N-T



5) Vysledky

rez	xi	yi	xe	ye	N	M	T	e = M/N
0	-0	0	-0	1.15	316.25	-90.921875	-142.74943	-0.2875
1	0.24694569	0.10326891	0.24694569	1.15	316.25	-73.797925	-128.23909	-0.23335312
2	0.49798112	0.19615515	0.49798112	1.15	316.25	-58.112853	-113.78758	-0.18375606
3	0.75266999	0.27849727	0.75266999	1.15	316.25	-43.990757	-99.396866	-0.13910121
4	1.0105696	0.35015215	1.0105696	1.15	316.25	-31.539556	-85.066011	-0.099729822
5	1.2712318	0.41099525	1.2712318	1.15	316.25	-20.851485	-70.791462	-0.065933551
6	1.5342034	0.46092083	1.5342034	1.15	316.25	-12.00358	-56.567399	-0.037955984
7	1.7990275	0.49984212	1.7990275	1.15	316.25	-5.0581413	-42.386077	-0.015994123
8	2.0652436	0.52769145	2.0652436	1.15	316.25	-0.063154368	-28.238191	-0.0001996976
9	2.3323893	0.54442044	2.3323893	1.15	316.25	2.9473679	-14.113258	0.0093197405
10	2.6	0.55	2.6	1.15	316.25	3.9531249	-4.5214321e-08	0.0125
11	2.8676107	0.54442044	2.8676107	1.15	316.25	2.9473679	14.113258	0.0093197405
12	3.1347564	0.52769145	3.1347564	1.15	316.25	-0.063154342	28.238191	-0.00019969752
13	3.4009725	0.49984212	3.4009725	1.15	316.25	-5.0581412	42.386077	-0.015994122
14	3.6657966	0.46092083	3.6657966	1.15	316.25	-12.00358	56.567399	-0.037955983
15	3.9287682	0.41099525	3.9287682	1.15	316.25	-20.851485	70.791462	-0.06593355
16	4.1894304	0.35015215	4.1894304	1.15	316.25	-31.539556	85.066011	-0.099729821
17	4.44733	0.27849727	4.44733	1.15	316.25	-43.990757	99.396866	-0.13910121
18	4.7020189	0.19615515	4.7020189	1.15	316.25	-58.112853	113.78758	-0.18375606
19	4.9530543	0.10326892	4.9530543	1.15	316.25	-73.797925	128.23909	-0.23335312
20	5.2	7.0038708e-10	5.2	1.15	316.25	-90.921875	142.74943	-0.2875

6) Report

ArchLBT prepared model
input = D:\Documents\2026\MASARCH2026\python\klenba02B_1_lbt.in

Geometry
span = 5.2
rise = 0.55
thickness D = 0.6
section depth min/max = 0.6 / 1.15
arch width b = 0.5
geom_CODE = 1
extrados mode = horizontal
blocks = 20
joints = 21

Loads
arch self-weight vertical = 49.345471
backfill vertical = 2.828800
variable vertical = 230.000000
total vertical = 282.174271

Toto je bez
násobku lamda -
jen zadání

Lower-bound skeleton
state = permanent + lambda * variable
solution = statically admissible reactions Rx, Ry, M0
constraints = $0 \leq N \leq f_c \cdot b \cdot D_i$ and $\text{abs}(M) \leq N \cdot (D_i/2 - N/(2 \cdot f_c \cdot b))$
shear constraint = $\text{abs}(T) \leq \mu \cdot N$
fc = 1100
b = 0.5
Nmax min/max = 330 / 632.5
mu = 0.6
status = optimal
lambda_max = 1.014455
governing joint = 10
governing M-N joints = 0, 10, 20
governing shear joints = 0, 20
governing compression joints = 10
governing tension joints = 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20
Rx = 316.250000
Ry = 142.749433
M0 = 90.921875

Prepared arrays
intrados/extrados = 21 x 4
local coordinates = 21 x 4
masonry loads = 21 x 3
backfill loads = 21 x 3
variable loads = 21 x 3

Global equilibrium check
moment origin = [0, 0]
loads sum Fx = 0.000000000e+00

6) Report / pokračovani

```
loads sum Fy = -2.854988653e+02
loads sum M = -7.422970496e+02
left reaction = [3.162500000e+02, 1.427494327e+02], M = 9.092187500e+01
right reaction = [-3.162500000e+02, 1.427494326e+02], M = -9.092187489e+01
residual sum Fx = 0.000000000e+00
residual sum Fy = 0.000000000e+00
residual sum M = 0.000000000e+00
```

Toto je přenásobeno
lamba - výsledné
maximalizované řešení

7) Generovana pole

```
local coordinates = 21 x 4
[[ 5.75000000e-01  1.57079633e+00  2.52286128e-01  2.06122734e-01]
 [ 5.23365543e-01  1.57079633e+00  2.55295422e-01  1.82937786e-01]
 [ 4.76922424e-01  1.57079633e+00  2.57995104e-01  1.60265969e-01]
 [ 4.35751366e-01  1.57079633e+00  2.60376315e-01  1.38036635e-01]
 [ 3.99923926e-01  1.57079633e+00  2.62431380e-01  1.16183139e-01]
 [ 3.69502373e-01  1.57079633e+00  2.64153780e-01  9.46422007e-02]
 [ 3.44539583e-01  1.57079633e+00  2.65538118e-01  7.33533254e-02]
 [ 3.25078942e-01  1.57079633e+00  2.66580106e-01  5.22582435e-02]
 [ 3.11154274e-01  1.57079633e+00  2.67276538e-01  3.13003883e-02]
 [ 3.02789781e-01  1.57079633e+00  2.67625285e-01  1.04243935e-02]
 [ 3.00000000e-01  1.57079633e+00  2.67625285e-01 -1.04243934e-02]
 [ 3.02789781e-01  1.57079633e+00  2.67276538e-01 -3.13003882e-02]
 [ 3.11154274e-01  1.57079633e+00  2.66580106e-01 -5.22582433e-02]
 [ 3.25078942e-01  1.57079633e+00  2.65538118e-01 -7.33533253e-02]
 [ 3.44539583e-01  1.57079633e+00  2.64153780e-01 -9.46422005e-02]
 [ 3.69502373e-01  1.57079633e+00  2.62431380e-01 -1.16183138e-01]
 [ 3.99923926e-01  1.57079633e+00  2.60376315e-01 -1.38036635e-01]
 [ 4.35751366e-01  1.57079633e+00  2.57995104e-01 -1.60265969e-01]
 [ 4.76922424e-01  1.57079633e+00  2.55295422e-01 -1.82937786e-01]
 [ 5.23365542e-01  1.57079633e+00  2.52286128e-01 -2.06122734e-01]
 [ 5.75000000e-01  1.57079633e+00  1.00000000e+00  4.48965309e-11]]
```

```
masonry loads = 21 x 3
[[0. 6.59105024 0.80106316]
 [0. 6.10191769 0.75404528]
 [0. 5.6484828 0.70848683]
 [0. 5.23714451 0.66567782]
 [0. 4.87361569 0.62681229]
 [0. 4.56287849 0.59296237]
 [0. 4.30914543 0.56505519]
 [0. 4.11582656 0.54385305]
 [0. 3.98550292 0.52993719]
 [0. 3.91990643 0.52369538]
 [0. 3.91990643 0.52531369]
 [0. 3.98550292 0.53477246]
 [0. 4.11582656 0.55184659]
 [0. 4.30914543 0.57611014]
 [0. 4.56287849 0.60694523]
 [0. 4.87361569 0.64355487]
 [0. 5.2371445 0.68497982]
 [0. 5.64848279 0.73011884]
 [0. 6.10191769 0.7775227]
 [0. 6.59105024 0.82656829]
 [0. 0. 0. ]]
```

```
backfill loads = 21 x 3
[[0. 0.26867691 0.0331743 ]
 [0. 0.27312655 0.03428222]
 [0. 0.27710148 0.03528733]
 [0. 0.2805948 0.03618265]
 [0. 0.28360042 0.03696195]
 [0. 0.28611313 0.03761982]
 [0. 0.28812856 0.03815169]
 [0. 0.28964321 0.03855385]
 [0. 0.29065444 0.03882353]
 [0. 0.29116049 0.03895884]
 [0. 0.29116049 0.03895884]
 [0. 0.29065444 0.03882353]
 [0. 0.28964321 0.03855385]
 [0. 0.28812856 0.03815169]
 [0. 0.28611313 0.03761982]
 [0. 0.28360042 0.03696195]
 [0. 0.2805948 0.03618265]
 [0. 0.27710148 0.03528733]
 [0. 0.27312655 0.03428222]
 [0. 0.26867691 0.0331743 ]
 [0. 0. 0. ]]
```

```
variable loads = 21 x 3
[[ 0. 10.92259777 1.34864422]
 [ 0. 11.10349035 1.39368476]
```

7) Generovana pole / pokracovani

[0.	11.26508441	1.43454578]
[0.	11.4070991	1.47094333]
[0.	11.52928758	1.50262447]
[0.	11.6314375	1.52936904]
[0.	11.7133713	1.5509912]
[0.	11.77494659	1.56734067]
[0.	11.81605633	1.57830385]
[0.	11.83662908	1.58380456]
[0.	11.83662908	1.58380456]
[0.	11.81605633	1.57830385]
[0.	11.77494659	1.56734067]
[0.	11.7133713	1.5509912]
[0.	11.6314375	1.52936904]
[0.	11.52928758	1.50262447]
[0.	11.4070991	1.47094333]
[0.	11.26508441	1.43454578]
[0.	11.10349035	1.39368476]
[0.	10.92259778	1.34864422]
[0.	0.	0.]]